

# YAO Pinshuo 姚品碩

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## RESEARCH INTERESTS

Protein–lipid recognition and its molecular logic; AI-driven and de novo protein design; high-throughput biophysical measurement; interpretable machine learning that turns large-scale binding data into general rules.

## EDUCATION

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024.10 – present | <b>Ph.D. Candidate, Chemistry and Biotechnology</b><br>Graduate School of Engineering, The University of Tokyo — Tsuboyama Laboratory, IIS<br>Advisor: Dr. Kotaro Tsuboyama. Large-scale mapping of protein–phosphoinositide recognition, combining high-throughput binding measurement with machine learning and AI-based protein design.                                                                                                            |
| 2024.04 – 2024.09 | <b>Research Student</b><br>Institute of Industrial Science, The University of Tokyo — Tsuboyama Laboratory<br>Joined the laboratory and established the lipid-binding screening pipeline.                                                                                                                                                                                                                                                             |
| 2022.04 – 2024.03 | <b>M.Sc., Life Sciences</b><br>Department of Interdisciplinary Sciences, Graduate School of Arts and Sciences, The University of Tokyo — Okada Laboratory (Laboratory of Pathology and Development), Institute for Quantitative Biosciences<br>Advisor: Prof. Yuki Okada. Thesis: <i>Establishment of an Ex Vivo Sperm Chromatin Manipulation Method via Divalent Cations</i><br>二価陽イオンを用いた精子クロマチンの体外改変方法の確立 — Outstanding Graduate Award (広域科学専攻奨励賞) |
| 2018.04 – 2022.03 | <b>B.Sc., Bioscience</b><br>Faculty of Life Sciences, Tokyo University of Agriculture — Laboratory of Functional Molecular Analysis (Yajima Laboratory)<br>Advisor: Prof. Shunsuke Yajima. Thesis: <i>Structural Analysis of LgnR, an IclR-Family Transcription Factor from Paracoccus sp. 43P</i><br>Paracoccus sp. 43P 由来の IclR ファミリー転写因子である LgnR の構造解析                                                                                             |
| 2016.10 – 2018.03 | <b>Japanese Language Program</b><br>Fuji International Language Institute, Tokyo                                                                                                                                                                                                                                                                                                                                                                      |

## RESEARCH POSITIONS

|                   |                                                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------|
| 2024.10 – present | <b>Research Fellow, JST SPRING-GX</b><br>Support for Pioneering Research Initiated by the Next Generation, The University of Tokyo |
| 2022.04 – 2024.03 | <b>Technical Assistant (技術補佐員)</b><br>Institute for Quantitative Biosciences (IQB), The University of Tokyo                        |

## PEER-REVIEWED PUBLICATIONS

- [1] Yao, P.; Nishimura, T.\*; Tsuboyama, K.\* Quantitative Kinetic Analysis of Protein–Phosphoinositide Binding by Optimized Liposome-Based Bio-Layer Interferometry. *Biochemistry* **2026**, special issue “Lipids and Lipidation”. DOI: 10.1021/acs.biochem.6c00344

Additional first-author manuscripts in preparation.

## CONFERENCE PRESENTATIONS

|         |                                                                                                                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025.01 | Decoding the Universal Principles of Protein–Phospholipid Binding<br><u>Yao, P.</u> ; Nishimura, T.; Tsuboyama, K. — IPR International Conference 2025, Awaji Island, Japan. Poster (international). |
| 2023.12 | Ex Vivo Sperm Chromatin Reconstitution: An Innovative Approach<br>IQB Research Exchange Meeting 2023, The University of Tokyo. Poster P-25.                                                          |

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|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023.06 | Establishment of an ex vivo Sperm Chromatin Reconstitution Method<br>16th Annual Meeting of the Japanese Society for Epigenetics, Hitotsubashi Hall, Tokyo. Poster P-49. |
| 2023.06 | Effects of Divalent Cations on Sperm Chromatin Structure and Ex Vivo Reconstitution<br>22nd UTokyo Life Sciences Symposium (BIO UT), Komaba Campus. Poster.              |
| 2022.11 | Analysis of Divalent Cation Effects on Sperm Chromatin Structure<br>Joint “Wakate-no-kai 2022”, Rinku, Osaka. Poster P-25.                                               |

## GRANTS AND FELLOWSHIPS

| TERM           | PROGRAMME            | PROJECT                                                                                                                                                                                                                                                          | OUTCOME                     |
|----------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| 2024 – present | <b>JST SPRING-GX</b> | Support for Pioneering Research Initiated by the Next Generation — Green Transformation Program for Advanced Human Resource Development, The University of Tokyo                                                                                                 | <b>AWARDED</b>              |
| FY2027         | <b>JSPS DC2</b>      | Elucidating the precise lipid-species recognition mechanism by large-scale analysis and machine learning                                                                                                                                                         | <b>UNDER REVIEW</b>         |
| 2026           | <b>SPReAD</b>        | MEXT “AI for Science” programme — Supporting Pioneering Research through AI for 1,000 Discovery challenges, 1st call. Predicting the lipid-binding specificity of lipid-binding domains and extracting recognition rules with a sequence–structure integrated AI | PASSED REVIEW,<br>NOT DRAWN |
| FY2026         | <b>JSPS DC2</b>      | Analysis of protein lifetime determinants by large-scale measurement and machine learning                                                                                                                                                                        | NOT AWARDED                 |
| FY2025         | <b>JSPS DC1</b>      | Analysis of protein lifetime determinants by large-scale measurement and machine learning                                                                                                                                                                        | NOT AWARDED                 |
| FY2024         | <b>JSPS DC1</b>      | Elucidating the significance of the sperm epigenome through ex vivo sperm reconstitution                                                                                                                                                                         | NOT AWARDED                 |

## AWARDS AND ACADEMIC ACTIVITIES

|         |                                                                                                                                                                                               |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024.03 | <b>Outstanding Graduate Award, Interdisciplinary Sciences (広域科学専攻奨励賞)</b><br>Graduate School of Arts and Sciences, The University of Tokyo, FY2023.                                           |
| 2023.08 | <b>Bio-SPM Summer School — collaborative project accepted</b><br>11th Bio-SPM Summer School, NanoLSI, Kanazawa University. High-speed AFM imaging of <i>in vitro</i> reconstituted complexes. |

## TECHNICAL SKILLS

|                 |                                                                                                                                                                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Biophysics      | Bio-layer interferometry (BLI) kinetics on liposome surfaces; liposome preparation of defined lipid composition; buffer optimization for protein–lipid interfaces.                                                                                                                                                              |
| High-throughput | Yeast surface display; pooled library construction and selection; next-generation sequencing and downstream analysis; fluorescence-activated cell sorting (FACS).                                                                                                                                                               |
| Protein work    | Recombinant expression in <i>E. coli</i> ; affinity, ion-exchange and size-exclusion chromatography; Western blot; site-directed mutagenesis and mutant series design.                                                                                                                                                          |
| Cell & imaging  | Fluorescence microscopy; intracytoplasmic sperm injection (ICSI) and mouse embryo handling; comet assay; chromatin decondensation–recondensation assays.                                                                                                                                                                        |
| Computational   | Structure prediction: AlphaFold2, AlphaFold3, RoseTTAFold, ESMFold2, Boltz-2, Chai-1. Generative design: RFdiffusion3, BoltzGen2, NISE, DISCO, ProteinMPNN, LigandMPNN. Protein language models: ESM-2, ESM3, ESM-C, SaProt. Surface and interface analysis: MaSIF. Machine learning on large-scale sequence–function datasets. |
| Model organisms | <i>Escherichia coli</i> , <i>Saccharomyces cerevisiae</i> , mouse.                                                                                                                                                                                                                                                              |
| Languages       | Chinese (native) · Japanese (advanced; full academic working proficiency, resident in Japan since 2016) · English (advanced; research, writing and presentation).                                                                                                                                                               |